

Leidy J. García Maza

Contact Information

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MSc in Pharmaceutical Sciences from the Universidade Federal do Rio Grande do Sul (UFRGS) in Porto Alegre, Brasil. Graduated in Chemistry from the Universidad del Atlántico in Barranquilla, Colombia. Hands-on experience in interdisciplinary research groups in the areas of Organic Chemistry, Analytical Chemistry, and Applied Microbiology. Also trained in advanced analytical techniques, including spectroscopic (one- and two-dimensional NMR, MS, FTIR, UV-Vis), chromatographic (HPLC-UV, TLC), and electrochemical (VC) methods, as well as data integration and visualization using R and Python.

Education

Master's in Pharmaceutical Sciences (early spring 2026), Universidade Federal do Rio Grande do Sul (UFRGS), Brasil.

Thesis: The biotechnological potential of the microbiota of sandbanks from the south coast of Rio Grande do Sul, Brazil: antibiofilm compounds and novel candidates for probiotics and postbiotics. **Advisor:** Alexandre José Macedo, PhD.

Bachelor's Degree in Chemistry (early spring 2023), Universidad del Atlántico, Barranquilla, Colombia.

Thesis: imino *Diels-Alder* reaction in the construction of several tetrahydroquinoline derivatives. Study of their properties as regulatory agents of Quorum sensing in *Pseudomonas aeruginosa*. **Advisor:** Carlos Meléndez Gómez, PhD.

Professional experience

HPLC Chemist Analyst (summer-winter 2023), Supramolecular Chemistry Research Group, Universidad del Atlántico, Colombia.

- Developed HPLC-UV analytical methodologies for quality control of active ingredients in medicinal products within the Atlántico Department, Colombia.

Advisor: Victoria Arana & Andrea Ramos. [Certificate - ES](#)

Academic Tutor (spring 2023), Educational Progress Route, Department of Atlántico, Colombia.

- Strengthened fundamental biology skills for upper secondary school students.
- Collaborated with Psycho-Pedagogical Matrices to deliver targeted educational programs.

Company: Psycho-Pedagogical Matrices. [Certificate - ES](#)

Research Intern (2021 - 2022), Organic and Biomedical Chemistry Research Group, Universidad del Atlántico, Colombia.

- Conducted organic synthesis employing conventional methodologies and established synthetic techniques.
- Separated and purified compounds using chromatographic techniques (thin-layer chromatography and column chromatography).
- Interpreted UV-Vis, IR, and NMR spectra (^1H , ^{13}C , DEPT 135, COSY, HMBC, HSQC, NOESY) and performed Mass Spectrometry (MS) analysis.

Advisor: Carlos Meléndez Gómez. [Certificate - ES](#)

Research Experience, Publications, and Projects

Peer-Reviewed Publications

- García, L.; Salgado, A.; Kouznetsov, V. V.; Meléndez, C.M., **Pyrrolo[2,1-a]isoquinoline scaffolds for developing anti-cancer agents.**, *RSC Adv*, **2024**, 14, 1710–1728. DOI: [10.1039/D3RA07047E](https://doi.org/10.1039/D3RA07047E).
- García Maza, L.J.; Orosco Flórez, D.F.; Salgado, A.R.M.; Rosales, W.; Mendoza-Torres, E.; Meléndez, C.M., **A mild catalyzed imino Diels-Alder reaction for the synthesis of N-(2-(o-tolyl)-1,2,3,4-tetrahydroquinoline-4-yl)formamide derivatives as regulators of Quorum Sensing in**

Conference Abstracts

- **Leidy Johana García Maza**, Rafaela Conceição de Souza, Alexandre José Macedo., **Marine bacterial extracts from Restinga Ecosystems as potential anti-biofilm agents against *Pseudomonas aeruginosa* via co-culture strategies.**, Expanded abstract published in the 10th Brazilian Conference on Natural Products (BCNP) and XXXVI Annual Meeting on Evolution, Systematics, and Micromolecular Ecology (RESEM), Belo Horizonte, Brazil, 2025.
- Rafaela Conceição de Souza, **Leidy Johana García Maza**, Alexandre José Macedo. **Evaluation of the probiotic potential of microorganisms isolated from coastal ecosystems of Rio Grande do Sul, Brazil.**, Expanded abstract published in the 10th Brazilian Conference on Natural Products (BCNP) and XXXVI Annual Meeting on Evolution, Systematics, and Micromolecular Ecology (RESEM), Belo Horizonte, Brazil, 2025.
- **García Maza, L.J.**; Orosco Flórez, D.F.; Salgado, A.R.M.; Rosales, W.; Mendoza-Torres, E.; Meléndez, C.M., **A mild catalyzed imino Diels-Alder reaction. Synthesis of N-(2-(o-tolyl)-1,2,3,4-tetrahydroquinoline-4-yl)formamide derivatives as antimicrobial agents.**, Expanded abstract published in the 35th Latin-American Chemistry Congress and 61st Brazilian Chemistry Congress, Rio de Janeiro, Brazil, 2022.
- Orosco Flórez, D.F.; **García Maza, L.J.**; Salgado, A.R.M.; Rosales, W.; Mendoza-Torres, E.; Meléndez, C.M., **Effect of a series of o-tolyl (1,2,3,4-tetrahydroquinolin-4-yl)formamide derivatives on bacterial growth and biofilm formation in *Pseudomonas aeruginosa*.**, Abstract published in the Caribbean Microbial Meeting - Memories 2022, Valledupar, Colombia, 2022.

Manuscripts in Preparation

- **García, L.**; Souza, R.; Da Rosa, D.; Beys da Silva, W.; Santi, L.; Schrank, A.; Henning Vainstein, M.; Macedo, A., **Marine derived bacteria from Restinga Ecosystems as a platform for antibiofilm agents and Next-Generation Probiotics and Postbiotics (NGPPs).**, Manuscript submitted to the journal Frontiers in Natural products (ID: 1844638). **2026.**
- **García, L.**; Da Rosa, D.; Beys da Silva, W.; Santi, L.; Schrank, A.; Henning Vainstein, M.; Macedo, A. **Short chain length polyhydroxyalkanoate (scl-PHA) produced by the marine bacterium *Pseudomonas sp.* MaRe06 inhibits *Staphylococcus aureus* biofilm formation.**, Manuscript will be submitted to the journal Bioorganic Chemistry. **2026.**
- **García, L.**; Souza, R.; Da Rosa, D.; Beys da Silva, W.; Santi, L.; Schrank, A.; Henning Vainstein, M.; Macedo, A. **Exploring probiotic related traits in the environmental bacterium *Pseudomonas sp.* MaRe06 through integrated phenotypic and genomic analysis.**, Manuscript will be submitted to the journal Food Science and Biotechnology. **2026.**

Honors and Awards

- **CAPES Scholarship, Brazil** (2024-2026). Fully funded postgraduate scholarship awarded for academic excellence and potential in research.
- **Short-Term Training Scholarship, Brazil** (September, 2025). "Biopolymers as Matrices for Pre/Probiotic Delivery". Fully funded by CNPq (National Council for Scientific and Technological Development, Brazil) and CABBIO (Latin American Center for Biotechnology).

Skills

Laboratory Techniques

- **Microbiology Assays:** Bioactive compound evaluation, bacterial growth kinetics, biofilm formation analysis, and microbial isolation, purification, and cultivation.
- **In Vivo Assays:** Trained in the handling and manipulation of *Galleria mellonella* and laboratory rat models, including basic in vivo experimental procedures.
- **Organic Synthesis:** Design and execution of synthetic routes, including compound identification using spectroscopic techniques.
- **Chromatographic Techniques:** HPLC, TLC, and column chromatography.
- **Spectroscopic and Spectrometric Analysis:** UV-Vis, IR, NMR (¹H, ¹³C, DEPT 135, COSY, HMBC, HSQC, NOESY), and Mass Spectrometry (MS).

Computational Skills

- **Programming and Data Analysis:** R, Python.
- **Scientific Software:** ChemDraw, SPSS, GraphPad Prism, MATLAB.
- **Data Management and Visualization:** Microsoft Excel.

Professional Credentials

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- **Professional Chemist License**
Issued by Professional Council of Chemistry Colombia (CPQCOL, by its acronym in Spanish).
License Number: PQ-09181.

Courses and certifications

- **Biopolymers as matrices for pre/probiotic release: Fundamentals, Applications and Entrepreneurship**
CABBIO (Latin American Center for Biotechnology) | Participant | 09/2025
Duration: 40 hours (60% practical and 40% theoretical) | Onsite: UNIARA, São Paulo, Brazil | Final Pitch 10.0/10 | Certificate - PT
- **Theoretical course - BIOMOL (PCR)**
MedReg 2025: Course on Regenerative Medicine and Tissue Engineering, Brazil | Assistant
08/2025 | Duration: 04 hours | Onsite: ICBS-UFRGS, Porto Alegre, Brazil | Certificate - PT
- **Theoretical Course - Flow Cytometry: Fundamentals and Applications**
MedReg 2025: Course on Regenerative Medicine and Tissue Engineering, Brazil | Assistant
08/2025 | Duration: 04 hours | Onsite: ICBS-UFRGS, Porto Alegre, Brazil | Certificate - PT
- **Applications of Recombinant DNA Technology: from Cancer to Fashion**
Escola Superior Instituto Butantan - São Paulo, Brazil | Assistant
05/2025 | Duration: 10 hours | Online | Certificate - PT
- **Uncomplicating the Biostatistics in SPSS: Fundamentals and Practical Applications for Postgraduates**
XVI Postgraduate Program Meeting in Pharmaceutical Sciences - Federal University of Rio Grande do Sul | Assistant
11/2024 | Duration: 04 hours | Onsite: ICBS-UFRGS, Porto Alegre, Brazil | Certificate - PT
- **Theoretical and Practical Course: Basic Aspects of Research with Laboratory Animals**
Centre for Reproduction and Experimentation of Laboratory Animals - Federal University of Rio Grande do Sul
06/2024 | Duration: 34 hours | Onsite: Campus do vale-UFRGS, Porto Alegre, Brazil | Certificate - PT
- **Advanced Course on High-Performance Liquid Chromatography (HPLC)**
Pharmaceutical Services & Consulting - México | Certified trainee
09/2023 | Duration: 16 hours | Final Score: 100/100 | Online | Certificate - ES
- **Bioinformatics Techniques for the Study of Proteins. Applications in Biomedical Research, Development, and Diagnostics**
Global Disease Research Colombia | Hands-on participant
08/2022 | Duration: 48 hours | Online | Certificate - ES

Conferences and Talks

Event: 17th Meeting of the Graduate Program in Pharmaceutical Sciences;

Type of participation: Graphical Abstract Presentation;

Title: *Scoping the unexplored marine bioactive compounds: Restinga-ecosystem bacterial metabolites as potential antibiofilm agents;*

Place: PORTO ALEGRE, BRAZIL - Universidade Federal do Rio Grande do Sul - Cultural Center;

Date: 26/11/2025 - 28/11/2025.

Event: 10th Brazilian Conference on Natural Products (BCNP);

Type of participation: Poster Presentation;

Place: BELO HORIZONTE, BRAZIL - Universidade Federal de Minas Gerais (UFMG);

Date: 04/11/2025 - 07/11/2025.

Certificate - EN

Event: Gaúcho Bioinformatics School;

Type of participation: Assistant;

Place: PORTO ALEGRE, BRAZIL - Brain Institute of Rio Grande do Sul - PUCRS;

Date: 21/07/2025 - 25/07/2025.

Certificate - PT

Event: Sustainable Drug Discovery (S-DISCO) Days;

Type of participation: Assistant;

Place: GDANSK, POLAND - Online;

Date: 19/05/2025 - 20/05/2025.

Certificate - EN

Event: 16th Meeting of the Graduate Program in Pharmaceutical Sciences;

Type of participation: Graphical Abstract Presentation;

Title: *Scoping the unexplored marine bioactive compounds: Restinga-ecosystem bacterial and fungal metabolites as potential antibiofilm agents;*

Place: PORTO ALEGRE, BRAZIL - Universidade Federal do Rio Grande do Sul - Cultural Center;

Date: 07/11/2024 - 09/11/2024.

Event: Cannabis Prospects;

Type of participation: Organizer;

Place: BARRANQUILLA, COLOMBIA - Universidad del Atlántico - Cultural Center;

Date: 16/11/2023.

Event: 35th Latin-American Chemistry Congress and 61st Brazilian Chemistry Congress;

Type of participation: Poster Presentation;

Title: *A mild catalyzed Imino-Diels Alder reaction. Synthesis of N-(2-(o-tolyl)-1,2,3,4-tetrahydroquinoline-4-yl)formamide derivatives as antimicrobial agents;*

Place: RIO DE JANEIRO, BRAZIL - Windsor Florida Convention Center;

Date: 14/11/2022 – 18/11/2022.

Certificate - PT

Event: Caribe Microbial Meeting;

Type of participation: Poster Presentation;

Title: *Effect of a series of N-(2-(o-tolyl)-1,2,3,4-tetrahydroquinoline-4-yl)formamide derivatives on bacterial growth and biofilm formation in Pseudomonas aeruginosa;*

Place: VALLEDUPAR, COLOMBIA - Universidad Popular del Cesar;

Date: 28/10/2022.

Certificate - ES

Event: 2nd Caribbean Chemistry Congress;

Type of participation: Assistant;

Place: BARRANQUILLA, COLOMBIA - Universidad del Atlántico;

Date: 27/02/2019 - 01/03/2019.

Certificate - ES

Volunteer Experience

- **STEM Pen Pal Volunteer - Letters to a Pre-Scientist Program**

Volunteer Mentor

2025 - Present | Remote

Year-long mentorship through handwritten letters with a middle school student to foster interest in STEM, sharing academic experiences and encouraging scientific curiosity.

- **Scout Member – Youth Leadership and Community Service**

Scouts Organization | Inactive Member

2012 – 2020 | Barranquilla, Colombia

Participated in community service activities, outdoor training, and teamwork-based challenges, developing leadership, responsibility, and problem-solving skills in collaborative environments.

Languages

- **Portuguese (Brazil):** Read: **C1** | Speak: **C1** | Write: **B1** | Listen: **C1**
- **English:** Read: **C1** | Speak: **C1** | Write: **C1** | Listen: **C1**
- **Spanish:** Native Speaker